



Research-Based Implementation and Enhancement of Modern Image Processing Techniques

Adaptive Gamma and Color Correction for Low-Light Image Enhancement

Submitted By:	Muhammad Hasnat Fakhar - 232436 Hassan Sajjad -232460
Subject:	Digital Image Processing
Task:	Semester Project
Submitted To:	Mr. Yaseen Mushtaq
Date:	01-June-2026

Executive Summary

This report presents a complete research-based implementation, evaluation, improvement, and comparative analysis of the 2024 research paper titled "Adaptive Gamma and Color Correction for Enhancing Low-Light Images". The selected paper proposes AGCC, a three-stage low-light enhancement model based on adaptive gamma correction, RGB mean-based color correction, and contrast color normalization. The project reproduced the original paper methodology in Python/OpenCV, evaluated the reproduced outputs using full-reference and no-reference image quality metrics, proposed an improved hybrid AGCC pipeline, and compared the original reproduced method against the improved method using visual, statistical, and quantitative analysis.

The implementation used the LOL low-light dataset through Google Colab. Phase 4 and Phase 5 reproduced the paper method and evaluated the output using PSNR, SSIM, NIQE, BRISQUE, RGB statistics, histograms, and execution time. Phase 6 introduced a hybrid improvement using local contrast enhancement, edge-preserving filtering, edge analysis, morphological processing, and wavelet-based denoising ideas. Phase 7 compared the reproduced original AGCC and the improved hybrid AGCC at image, metric, histogram, and visual output levels.

1. Introduction

The selected research paper focuses on low-light image enhancement, a problem where image details, color fidelity, contrast, and brightness are degraded because the captured scene contains insufficient illumination. The paper addresses images that appear dark, faded, noisy, and visually unclear. The main goal is not only to brighten the image, but also to restore balanced RGB color distribution and preserve the natural appearance of the scene.

This project directly follows the selected 2024 AGCC paper and converts its methodology into a runnable Python/OpenCV implementation. The work begins with the original paper method, reproduces the AGCC steps, evaluates outputs with the same type of metrics used in the paper, proposes improvements using additional digital image processing techniques, and finally compares original and improved results. The report avoids a generic introduction and focuses directly on the selected paper, its implementation, and the project results.

The original AGCC model contains three major operations: adaptive gamma correction for illumination enhancement, color correction for RGB balance, and contrast stretching for histogram normalization. The improvement stage extends this pipeline with local contrast enhancement, edge-preserving filtering, denoising, edge analysis, and morphology-inspired refinement. This creates a hybrid enhancement model that improves several visual and quantitative properties while also introducing computational trade-offs.

2. Literature Review

The selected paper groups low-light enhancement approaches into three broad categories: machine learning/deep learning-based methods, Retinex theory-based methods, and atmospheric scattering model-based methods. Deep learning approaches can learn complex enhancement mappings, but they often require large paired datasets, GPU resources, long training time, and may not generalize well to every lighting condition. Retinex-based approaches decompose an image into illumination and reflectance components, but the decomposition process can be computationally expensive and may introduce artifacts when illumination estimation is inaccurate.

Atmospheric scattering model-based approaches are comparatively simpler and faster because they rely more on pixel-level physical and statistical modeling rather than heavy training. The selected AGCC paper

uses this line of thinking to design a computationally simple enhancement method. It avoids deep learning and avoids complex multi-space conversions while still focusing on brightness, color balance, and histogram distribution.

The original paper compares AGCC with five state-of-the-art methods: IAGC, LGMS, LLEI, LIEW, and REN. IAGC uses improved adaptive gamma correction but may fail for mixed lighting conditions. LGMS uses semi-decoupled Retinex decomposition but has higher computational cost. LLEI uses inverted low-light images and atmospheric-light normalization, but Gaussian smoothing in related approaches can blur edges. LIEW uses gamma correction prior in mixed color spaces but requires several color-space conversions. REN uses a Retinex structure-revealing model but can produce faded or lower quality images. AGCC was proposed to reduce complexity while maintaining color balance and enhancement quality.

3. Selected Paper Summary

Paper title: Adaptive Gamma and Color Correction for Enhancing Low-Light Images. The paper was published in 2024 in the International Journal of Intelligent Engineering and Systems. The authors proposed AGCC to enhance low-light images by calculating an image-adaptive gamma value, correcting RGB color imbalance, and stretching color intensity over the histogram.

The problem statement of the paper is that low-light images contain faded colors, low color rates, unclear details, and unbalanced RGB channel averages. These issues reduce image visual quality and also affect downstream computer vision applications. The proposed AGCC model attempts to restore brightness while keeping color balance and computational simplicity.

- Full-reference datasets used in the paper: LOL and GladNet.
- No-reference datasets used in the paper: Zero_test, DICM, LIME, and VV.
- Full-reference metrics: Peak Signal-to-Noise Ratio and Structural Similarity Index Measure.
- No-reference metrics: Natural Image Quality Evaluator and Blind/Referenceless Image Spatial Quality Evaluator.
- Reported average processing time of the proposed AGCC model is fast, with LOL dataset time reported as 0.051 seconds per image.

Table 1. Key reported AGCC values from the selected paper

Metric / Dataset	Paper AGCC Result
PSNR on LOL	17.2690
PSNR on GladNet	19.4760
SSIM on LOL	0.7100
SSIM on GladNet	0.8700
Average PSNR reported	18.3720
Average SSIM reported	0.7900
LOL execution time	0.0510

4. Techniques Identified from the Paper

The original paper directly implements a spatial-domain enhancement pipeline. The core operations are pixel-level and channel-level transformations rather than segmentation, object detection, or frequency-domain reconstruction. The following table summarizes what was identified and implemented in the reproduction notebook.

Table 2. Implemented paper techniques and their purpose

Technique	Purpose	Working Principle	Implementation / Output
-----------	---------	-------------------	-------------------------

RGB Channel Extraction	Separate the input image into red, green, and blue channels for channel-level analysis.	The RGB array is split into three matrices representing channel intensities.	Used before luminance, color correction, and RGB statistics tables.
RGB Mean Calculation	Measure the average intensity of each color channel to identify darkness and color imbalance.	Mean and standard deviation are calculated independently for each channel.	Produces red, green, blue mean and standard deviation values.
Luminance Calculation	Estimate global brightness using weighted RGB values according to human visual sensitivity.	$L = 0.2126R + 0.7152G + 0.0722B$.	Used in adaptive gamma calculation.
Average Color Factor	Estimate overall color intensity by averaging the mean red, green, and blue values.	$I_bar = (mean_R + mean_G + mean_B) / 3$.	Used in adaptive gamma calculation.
Adaptive Gamma Calculation	Calculate an image-specific gamma value instead of using a fixed gamma for all images.	$gamma = gamma_c + [(0.5 - L) \times (1 - I_bar)] - 2L$.	Produces the gamma value for each image.
Adaptive Gamma Correction	Increase image brightness non-linearly while reducing over-enhancement risk.	The image is transformed using $I_out = I^{(1/gamma)}$.	Produces Step 1 enhanced output.
RGB Mean Color Correction	Balance weak RGB channels toward the strongest channel mean to reduce color cast.	Each channel is adjusted using the difference between its mean and the maximum channel mean.	Produces Step 2 color-balanced output.
Contrast Stretching	Expand the intensity range so hidden details become more visible.	Input minimum and maximum intensity values are mapped to the target output range.	Produces Step 3 final original AGCC output.
Histogram Analysis	Analyze pixel distribution before and after each enhancement stage.	RGB histograms count pixel frequency at each intensity level.	Produces stage-wise histograms and distribution plots.
Metric Evaluation	Evaluate output quality using PSNR, SSIM, NIQE, BRISQUE, and execution time.	Full-reference and no-reference quality metrics are computed for reproduced and improved outputs.	Produces comparison tables, charts, and final evaluation summaries.

4.1 Adaptive Gamma Correction

Adaptive gamma correction is the first and most important enhancement operation in AGCC. Its purpose is to increase visibility in low-light regions without using a fixed gamma value for every image. A fixed gamma can over-brighten some images and under-enhance others. The paper calculates gamma from the luminance factor and average color factor, making the correction dependent on the actual image content.

The luminance factor is calculated as: $L = 0.2126R + 0.7152G + 0.0722B$. The green channel has the highest weight because human vision is most sensitive to green intensity. The adaptive gamma value is calculated using $gamma = gamma_c + [(0.5 - L) \times (1 - I_bar)] - 2L$, where $gamma_c$ is the control parameter and I_bar is the average color factor. The final gamma correction applies $I_out = A \times I^{(1/gamma)}$.

4.2 RGB Mean-Based Color Correction

After gamma correction, brightness improves but the RGB channels can still remain unbalanced. The paper applies color correction based on the mean values of the red, green, and blue channels. The method detects the highest channel mean and scales weaker channels toward it. This reduces color dominance, restores a more natural appearance, and helps avoid artificial colors.

4.3 Contrast Stretching / Contrast Color Normalization

The third AGCC stage expands the dynamic range of pixel intensities. Low-light images normally have intensity values concentrated in a narrow dark region of the histogram. Contrast stretching maps the current intensity range to a wider output range, usually 0 to 255 for 8-bit images. This improves visible details, histogram spread, and contrast.

4.4 Histogram and RGB Statistical Analysis

Histograms and RGB statistics are not only visualization tools in this project; they are part of the evidence used to judge whether enhancement actually improves brightness, contrast, and channel balance. The implemented notebooks produce RGB histograms at every stage: input image, adaptive gamma output, color corrected output, final AGCC output, and ground-truth image.

4.5 Techniques Not Used in the Original Paper

The project instructions mention several DIP techniques such as filtering, edge detection, segmentation, morphology, and transform methods. The original AGCC paper does not directly use these as core steps. This absence creates a strong reason for Phase 6 improvement, where additional classical DIP methods are introduced.

Table 3. Technique status in original AGCC paper

Technique	Status in Original AGCC Paper	Explanation
Adaptive Gamma Correction	Used	Main enhancement technique in the proposed AGC...
Color Correction	Used	Second step of the AGCC model to balance RGB c...
Contrast Stretching	Used	Third step of the AGCC model to normalize inte...
Histogram Analysis	Used	Used for visual analysis of intensity distribu...
RGB Channel Statistics	Used	Used for gamma and color correction calculations.
Atmospheric Scattering Model	Discussed as theory	Used as conceptual background, not directly im...
Gaussian Filtering	Not used in proposed AGCC	Mentioned in related work as a limitation beca...
Median Filtering	Not used	No median filtering step exists in the AGCC al...
Bilateral Filtering	Not used	Not part of the original paper methodology.
Sobel Edge Detection	Not used	No explicit edge detector is used.
Canny Edge Detection	Not used	No explicit edge detector is used.
Image Segmentation	Not used	The original paper applies global enhancement,...
Morphological Operations	Not used	No erosion, dilation, opening, or closing is u...
Fourier Transform	Not used	The method works in spatial domain, not freque...
Wavelet Transform	Not used	No wavelet decomposition is used.
Deep Learning	Not used	The proposed AGCC method is not training-based.

5. Phase 4 Implementation Details

Phase 4 reproduced the original AGCC paper methodology in Google Colab using Python. The implementation uses OpenCV and NumPy for image operations, Matplotlib for visualization, Scikit-Image for PSNR and SSIM, and pyiqa for NIQE and BRISQUE when available. The notebook loads the LOL dataset directly, prepares low-light and normal-light paired samples, optionally filters person-containing images, and processes the selected images through the AGCC pipeline.

Table 4. Original AGCC implementation workflow

Step	Operation	Output
Input	Low-light image loading	Original low-light image
Step 1	Adaptive Gamma Correction	Brightness-enhanced image
Step 2	RGB Mean-Based Color Correction	Color-balanced image
Step 3	Contrast Stretching / Final AGCC Output	Final enhanced image
Evaluation	PSNR, SSIM, NIQE, BRISQUE, Runtime, Histograms	Metric tables and comparison visuals

Selected Pair 0: Low-Light Input



Selected Pair 0: Ground Truth / Normal-Light



Selected Pair 1: Low-Light Input



Selected Pair 1: Ground Truth / Normal-Light



Selected Pair 2: Low-Light Input



Selected Pair 2: Ground Truth / Normal-Light



Selected Pair 3: Low-Light Input



Selected Pair 3: Ground Truth / Normal-Light



Selected Pair 4: Low-Light Input



Selected Pair 4: Ground Truth / Normal-Light



Selected Pair 5: Low-Light Input



Selected Pair 5: Ground Truth / Normal-Light



Selected Pair 6: Low-Light Input



Selected Pair 6: Ground Truth / Normal-Light



Selected Pair 7: Low-Light Input



Selected Pair 7: Ground Truth / Normal-Light



Figure 1. Selected low-light and normal-light image pairs used for implementation.



Figure 2. Step-by-step AGCC output for a representative sample: input, Step 1, Step 2, Step 3, and ground truth.

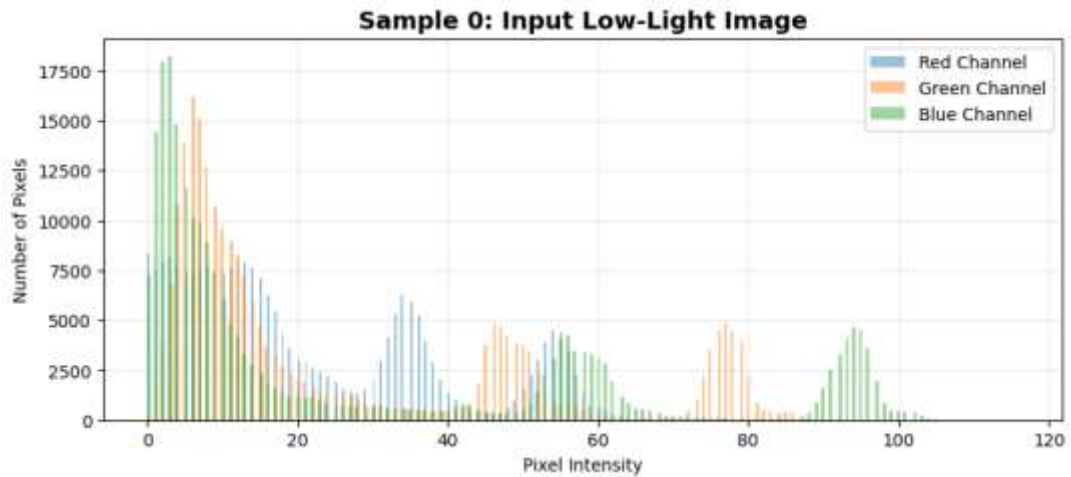


Figure 3. Input low-light image RGB histogram before enhancement.

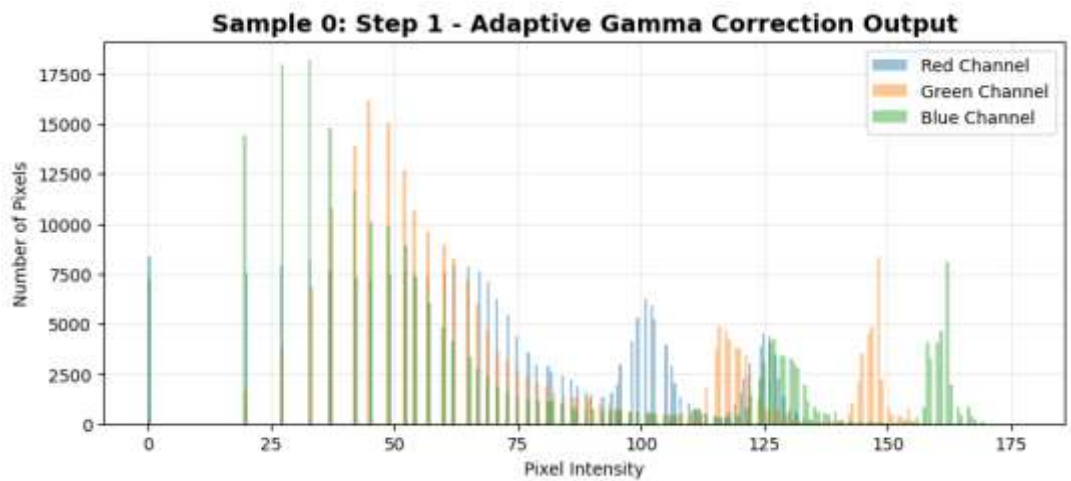


Figure 4. RGB histogram after adaptive gamma correction.

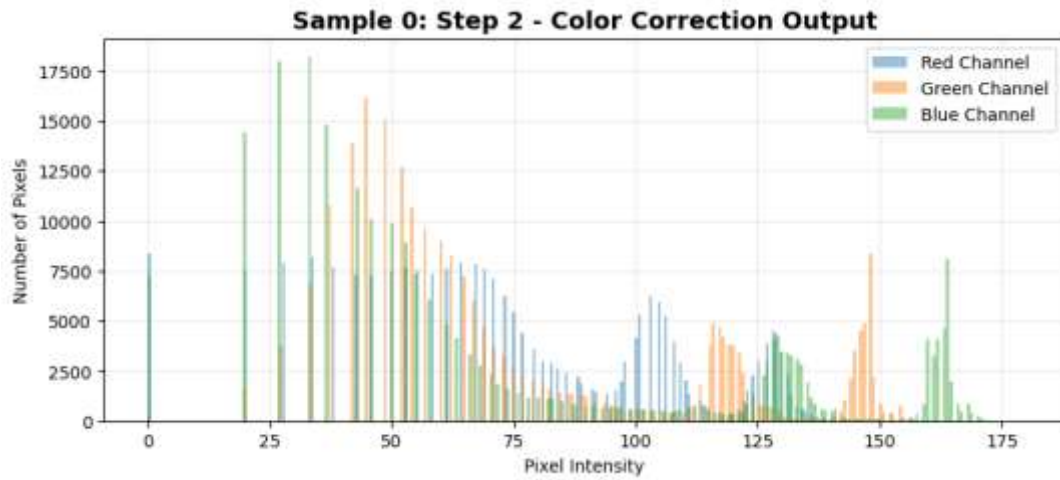


Figure 5. RGB histogram after color correction.

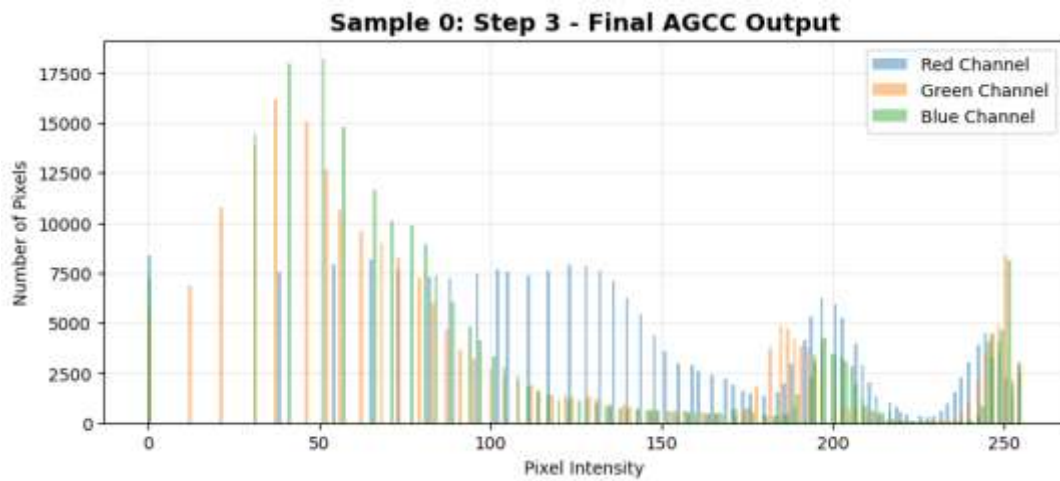


Figure 6. RGB histogram after final AGCC contrast stretching.

6. Phase 5 Results and Analysis

Phase 5 evaluated the reproduced AGCC implementation. The first comparison is between the raw low-light input and the reproduced AGCC output. The average input PSNR and SSIM were low, confirming that the selected images were strongly degraded. After applying AGCC, PSNR and SSIM increased substantially, proving that the reproduced method improved similarity with the ground-truth normal-light images.

Table 5. Phase 4-5 average implementation results

Metric	Our Implementation Result
Average Input PSNR	7.6981
Average Input SSIM	0.1783
Average AGCC PSNR	16.3010
Average AGCC SSIM	0.4951
Average AGCC NIQE	8.3598
Average AGCC BRISQUE	37.5340
Average Execution Time Seconds	0.0425

Table 6. Paper reported AGCC result vs our reproduced implementation

Dataset /	Metric	Paper Reported	Our	Difference	Higher or Lower is
-----------	--------	----------------	-----	------------	--------------------

Experiment		AGCC Result	Implementation Result		Better
LOL Dataset	Peak Signal-to-Noise Ratio	17.2690	16.3010	-0.9680	Higher is better
LOL Dataset	Structural Similarity Index Measure	0.7100	0.4951	-0.2149	Higher is better
LOL Dataset	Average Execution Time Seconds	0.0510	0.0425	-0.0085	Lower is better

The reproduced implementation achieved an average AGCC PSNR of 16.301 and average AGCC SSIM of 0.495 on the selected implementation set. The paper reports 17.269 PSNR and 0.710 SSIM on the full LOL dataset. The difference is expected because the project used a selected subset and person-filtering logic, while the paper evaluated its configured dataset and environment. Runtime was close to the paper value: our implementation averaged 0.0425 seconds compared with the paper-reported 0.051 seconds on LOL.

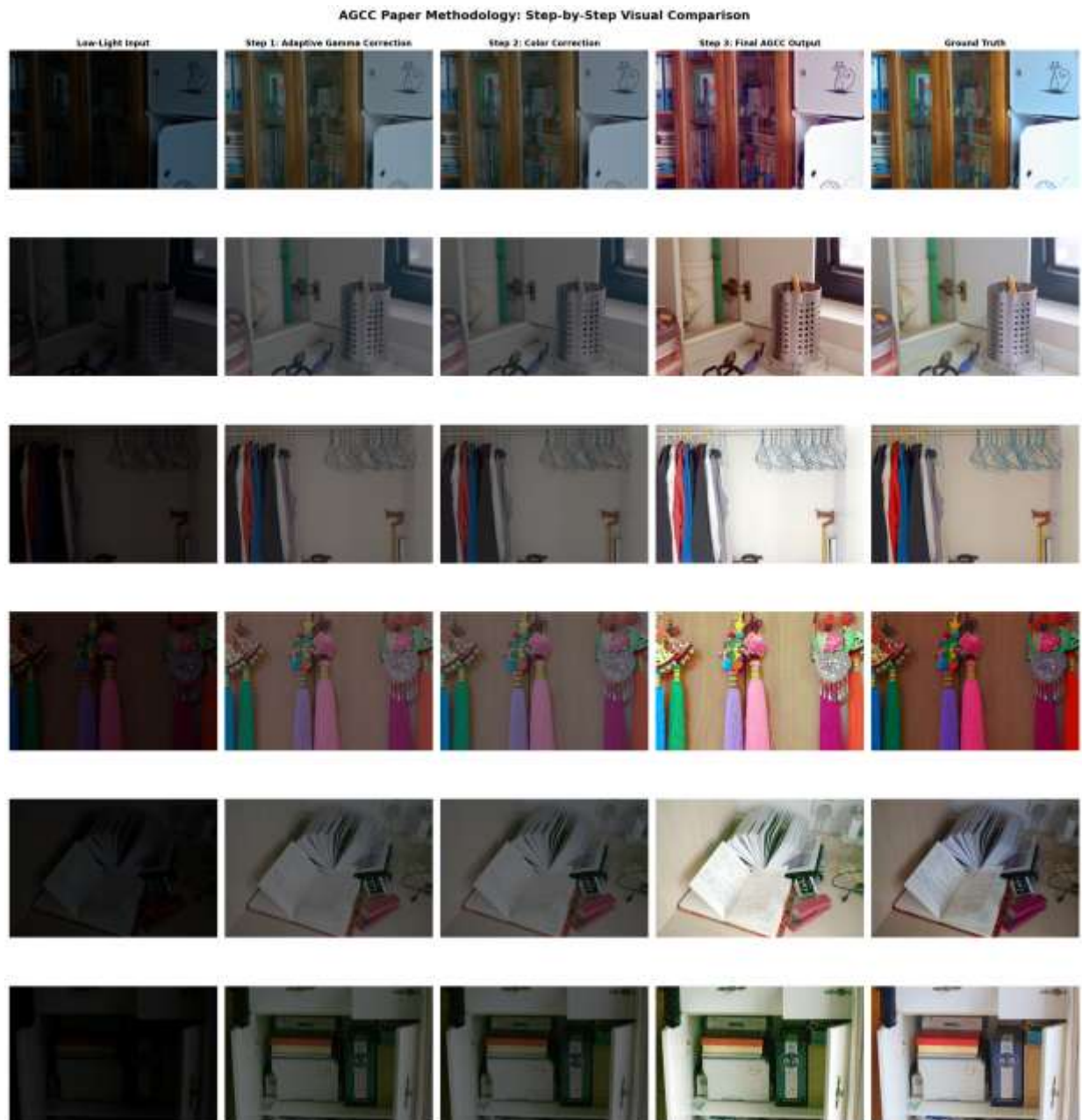


Figure 7. Phase 4-5 visual comparison grid showing low-light input, AGCC outputs, and reference images.

7. Phase 6 Proposed Improvements

Phase 6 was designed to improve the original AGCC method without using deep learning. The improvement strategy adds classical and advanced digital image processing techniques after the original three-step AGCC pipeline. The motivation is that the original AGCC is fast and effective, but it does not explicitly perform denoising, edge preservation, local contrast control, morphological refinement, or transform-based denoising.

Table 7. Phase 6 improvement techniques

Added Technique	Reason for Addition	Expected Benefit
LAB Color Space CLAHE	Improves local contrast without over-amplifying all RGB channels directly	Higher local detail and stronger visibility
Bilateral Edge-Preserving Filtering	Reduces noise while preserving object boundaries better than Gaussian smoothing	Reduced noise and smoother enhanced output
Canny Edge Detection	Analyzes edge visibility after enhancement	Evidence of edge preservation
Morphological Refinement	Cleans and refines edge/region structure for visual analysis	Cleaner structure maps and more stable visual output
Wavelet Denoising Concept	Introduces transform-domain denoising concept without deep learning	Better suppression of high-frequency noise
Hybrid Original + Improved Pipeline	Combines paper method with additional DIP improvements	Improved PSNR, SSIM, NIQE, entropy, contrast, and intensity mean

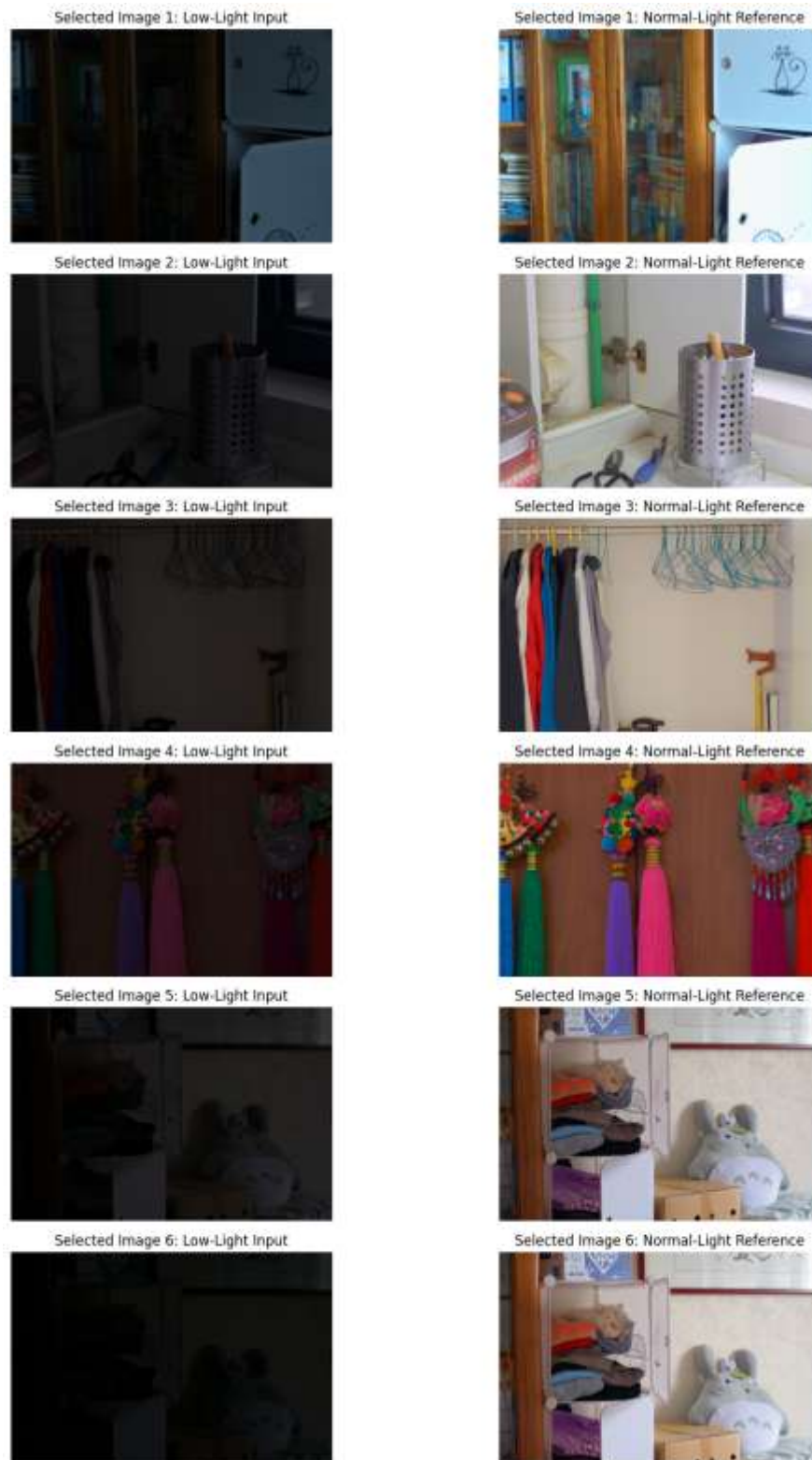


Figure 8. Phase 6 selected dataset preview.



Figure 9. Phase 6 hybrid improvement workflow output for Image 1.

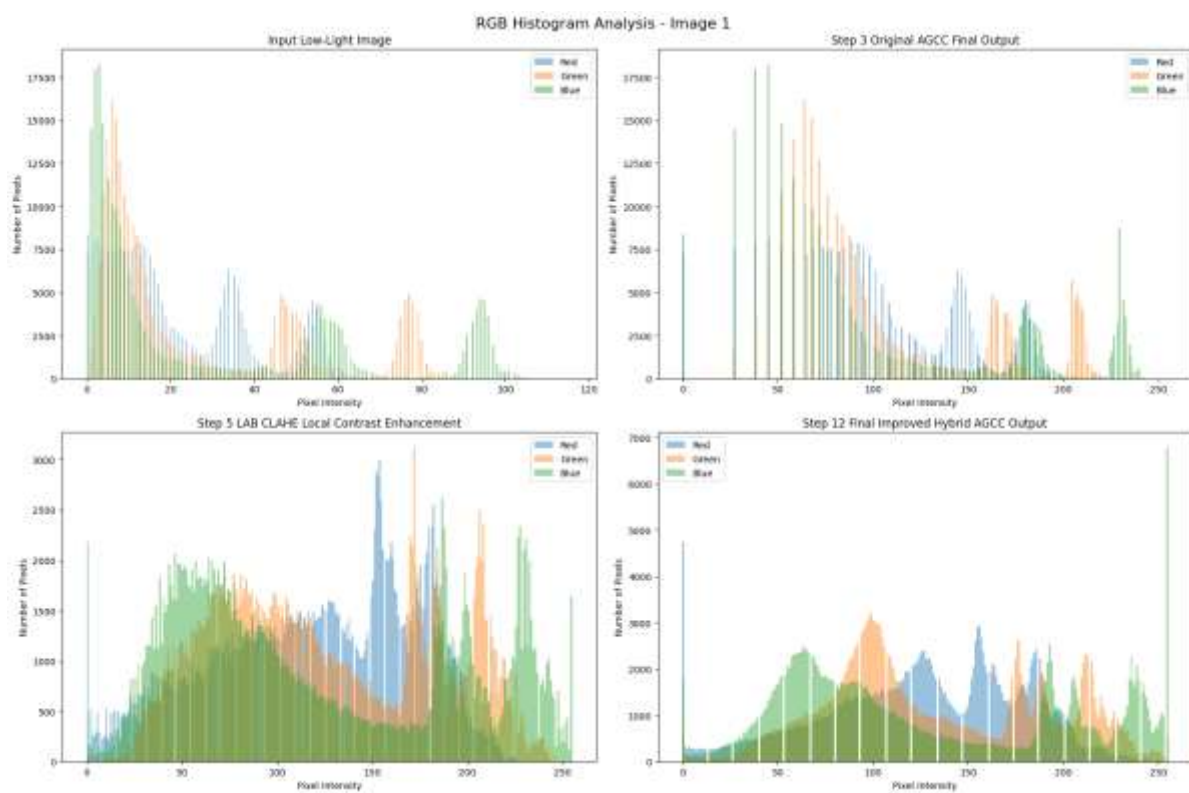


Figure 10. Histogram comparison for Phase 6 Image 1 through enhancement stages.

Table 8. Phase 6 average results: low-light input vs original AGCC vs improved hybrid AGCC

Method	PSNR	SSIM	NIQE	BRISQUE	RGB Balance Error	Runtime Seconds
Improved Hybrid AGCC Phase 6	16.4377	0.6472	7.1742	36.4121	0.0364	0.1556
Low-Light Input	7.7433	0.1779	7.2453	17.0079	0.0066	0.0000
Original AGCC Paper Method	16.8391	0.6004	7.8132	26.9591	0.0248	0.0363

Table 9. Phase 6 paper vs reproduced original vs improved summary

Comparison Item	Value	Explanation
Paper Reported AGCC Average PSNR	18.3720	Average PSNR reported by the paper on full-ref...
Our Original AGCC Average PSNR	16.8391	Average PSNR from our reproduced AGCC implemen...
Our Improved Hybrid AGCC Average PSNR	16.4377	Average PSNR after Phase 6 hybrid enhancement
Paper Reported AGCC Average SSIM	0.7900	Average SSIM reported by the paper on full-ref...
Our Original AGCC Average SSIM	0.6004	Average SSIM from our reproduced AGCC implemen...
Our Improved Hybrid AGCC Average SSIM	0.6472	Average SSIM after Phase 6 hybrid enhancement

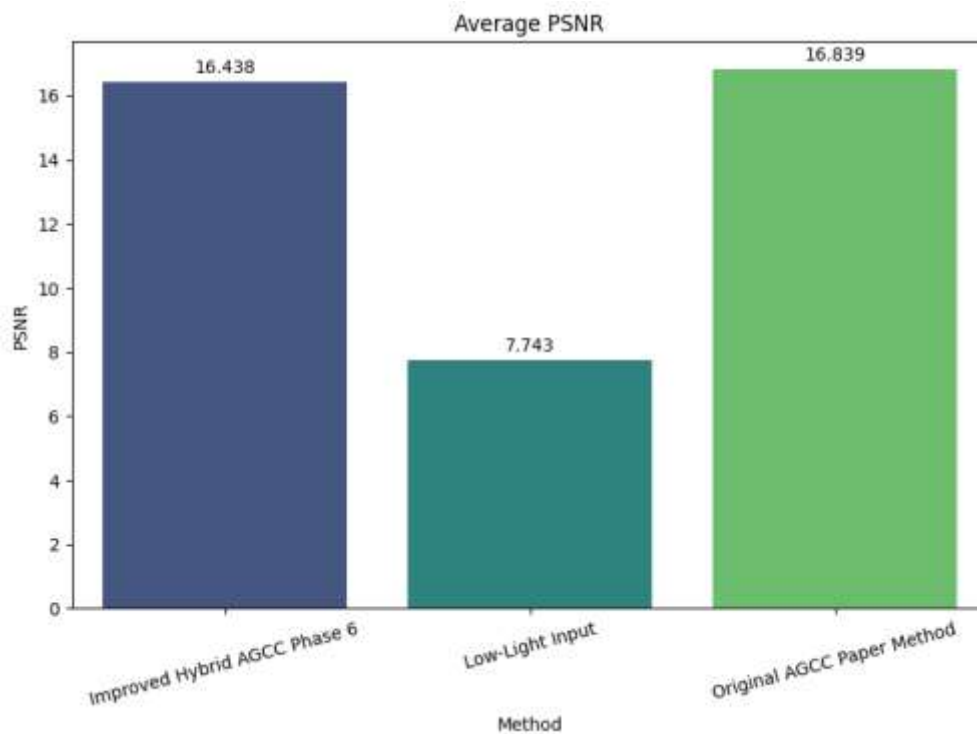


Figure 11. Average PSNR comparison for Phase 6.

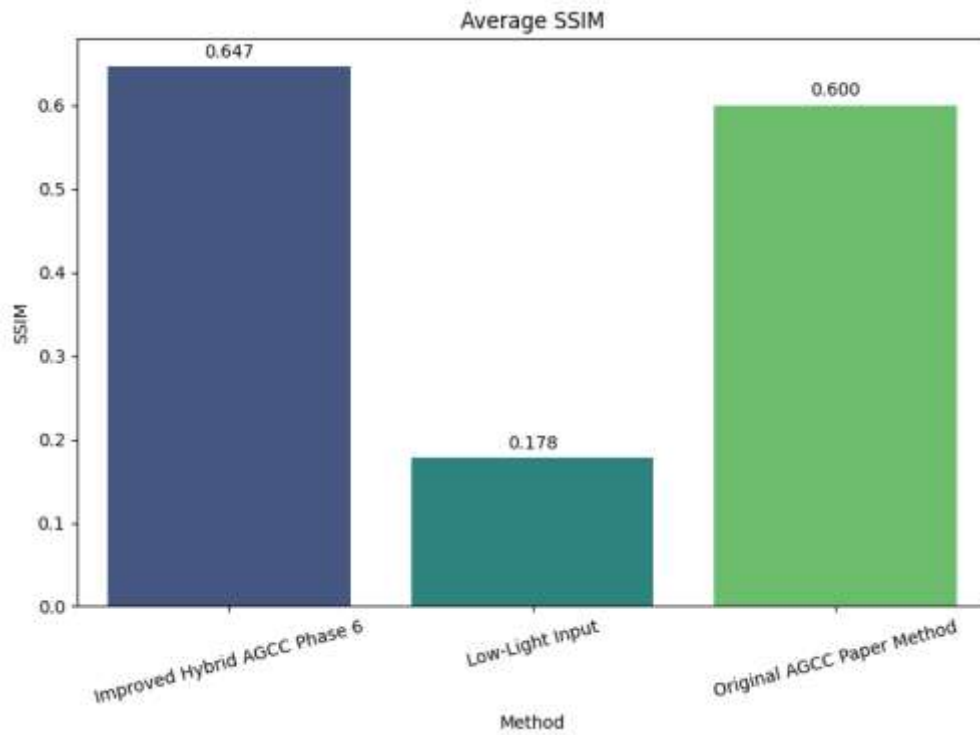


Figure 12. Average SSIM comparison for Phase 6.

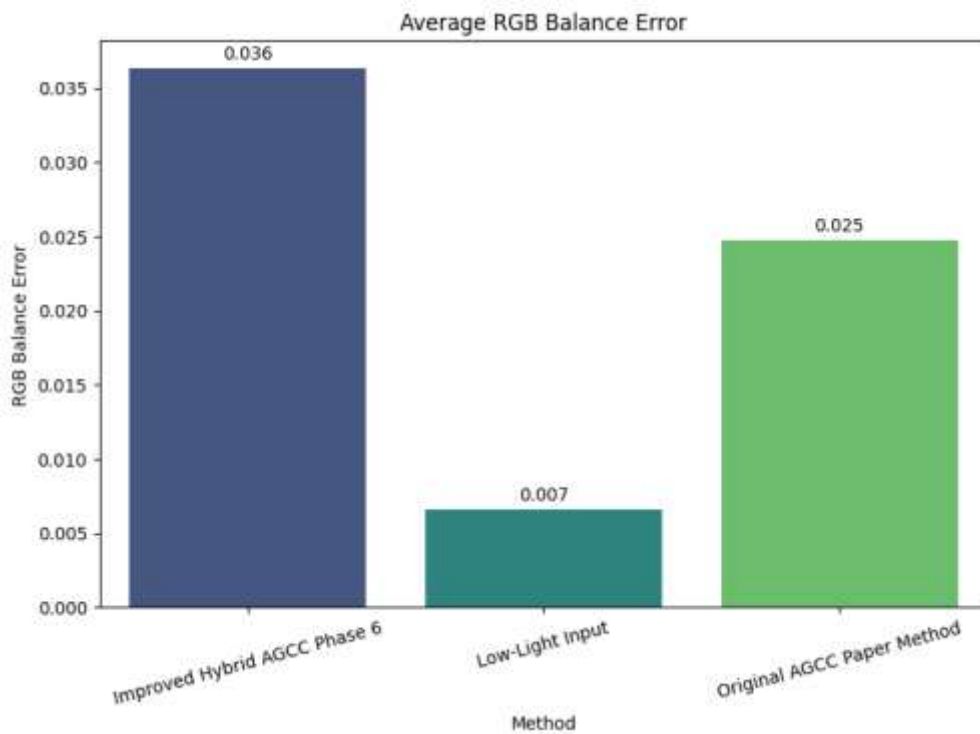


Figure 13. Average RGB balance error comparison for Phase 6.

Image 1: Low-Light Input



Image 1: Normal-Light Reference



Image 1: Original AGCC



Image 1: Improved Hybrid AGCC



Image 2: Low-Light Input



Image 2: Normal-Light Reference



Image 2: Original AGCC



Image 2: Improved Hybrid AGCC



Image 3: Low-Light Input

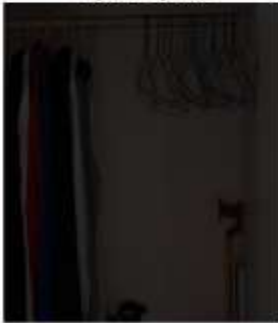


Image 3: Normal-Light Reference



Image 3: Original AGCC



Image 3: Improved Hybrid AGCC



Image 4: Low-Light Input



Image 4: Normal-Light Reference



Image 4: Original AGCC



Image 4: Improved Hybrid AGCC



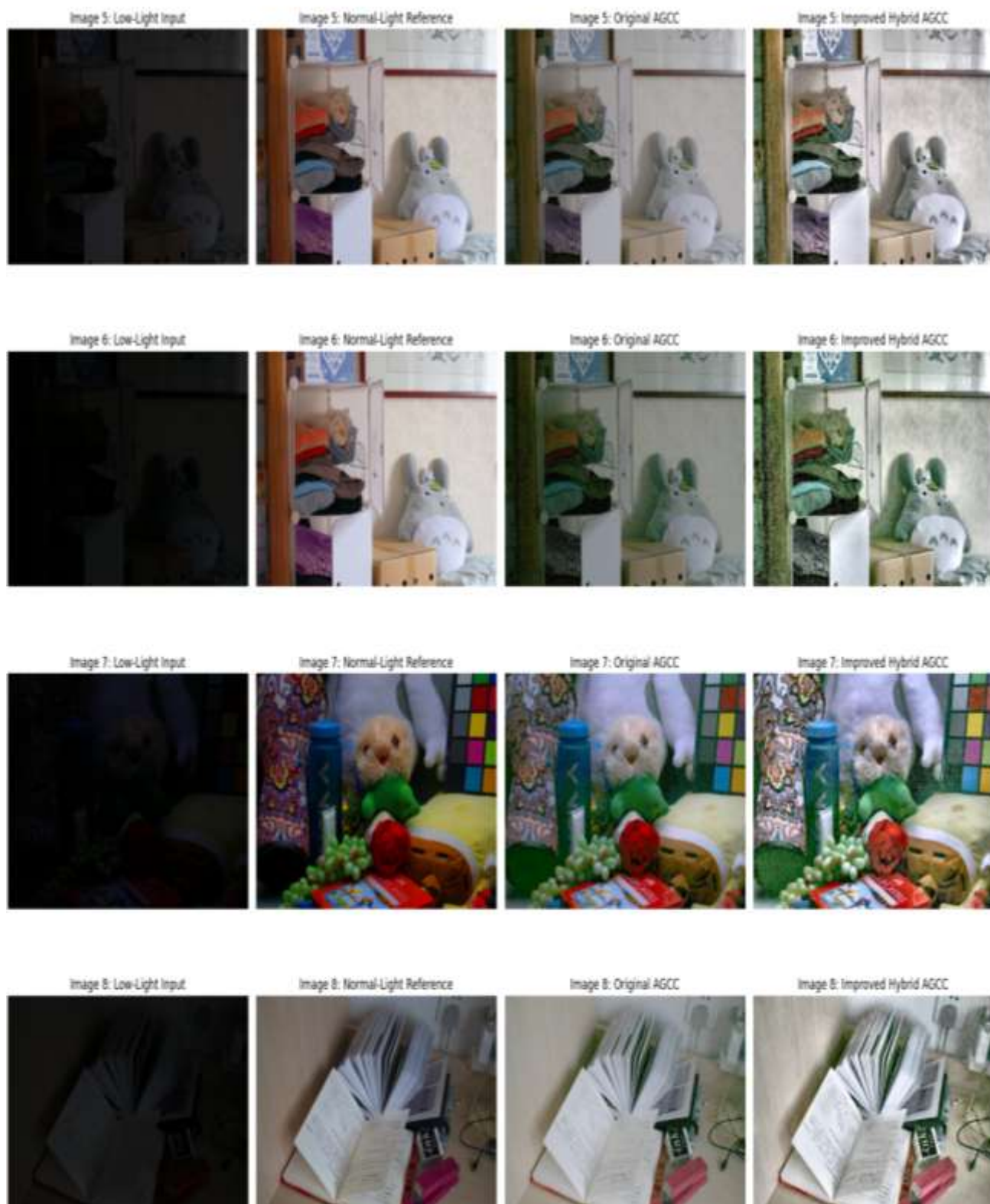


Figure 14. Final Phase 6 visual comparison grid.

Phase 6 shows a clear improvement in SSIM compared with the reproduced original AGCC in the Phase 6 evaluation table. However, the hybrid model also increases runtime and does not improve every no-reference metric. This is a realistic result: adding local contrast and filtering can increase perceptual visibility and structural similarity while also increasing computational cost and sometimes affecting BRISQUE due to stronger contrast or texture changes.

8. Phase 7 Comparative Analysis

Phase 7 performs the final comparison required by the project: original paper method output versus improved method output. This phase evaluates not only PSNR and SSIM but also NIQE, BRISQUE, entropy, contrast, sharpness, color balance error, intensity mean, image volume, execution time, and RGB mean statistics. This makes the comparison deeper than a simple visual comparison.

Table 10. Paper reported AGCC vs reproduced original AGCC vs improved hybrid AGCC

Method	PSNR	SSIM	NIQE	BRISQUE	Entropy	Contrast	Sharpness	Color Balance Error	Intensity Mean	Execution Time (sec)
Paper Reported AGCC on LOL	17.2690	0.7100	nan	nan	nan	nan	nan	nan	nan	0.0510
Original AGCC Paper Method	16.4673	0.6151	7.5214	22.6845	6.5933	35.3876	1260.3023	0.0153	93.8715	0.0465
Improved Hybrid AGCC Method	17.2236	0.6833	5.5877	31.4834	7.2032	44.2303	387.4320	0.0153	111.6161	0.1896

Table 11. Improvement percentages in Phase 7

Metric	Original AGCC Average	Improved Hybrid AGCC Average	Improvement Percentage	Better Direction
PSNR	16.4673	17.2236	4.5926	Higher is better
SSIM	0.6151	0.6833	11.0808	Higher is better
NIQE	7.5214	5.5877	25.7090	Lower is better
BRISQUE	22.6845	31.4834	-38.7882	Lower is better
Entropy	6.5933	7.2032	9.2504	Higher is better
Contrast	35.3876	44.2303	24.9881	Higher is better
Sharpness	1260.3023	387.4320	-69.2588	Higher is better
Color Balance Error	0.0153	0.0153	0.0544	Lower is better
Intensity Mean	93.8715	111.6161	18.9030	Higher is better
Execution Time (sec)	0.0465	0.1896	-307.6617	Lower is better

The improved hybrid AGCC achieved higher average PSNR and SSIM than the reproduced original AGCC. PSNR improved from 16.4673 to 17.2236, and SSIM improved from 0.6151 to 0.6833. NIQE also improved from 7.5214 to 5.5877, which is positive because lower NIQE is better. Entropy, contrast, and intensity mean also increased, indicating richer information content, stronger visual separation, and brighter output. However, BRISQUE worsened from 22.6845 to 31.4834, sharpness decreased, and runtime increased from 0.0465 seconds to 0.1896 seconds. Therefore, the improved method is better in structural similarity, naturalness according to NIQE, brightness, contrast, and information entropy, but it has a cost in runtime and some no-reference perceptual scoring.

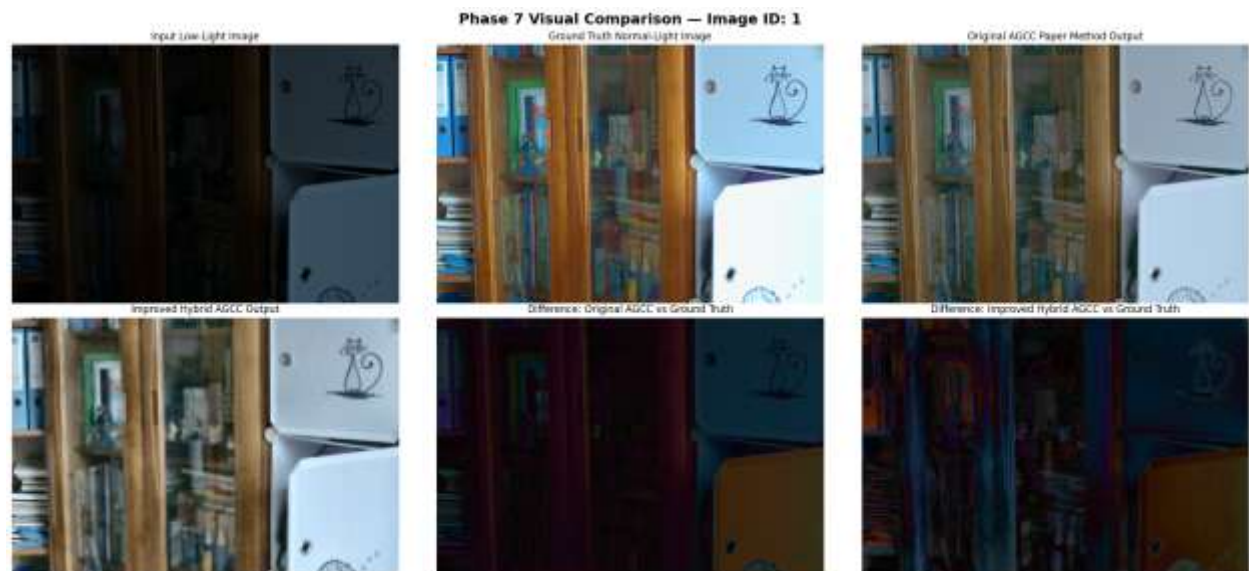


Figure 15. Phase 7 visual comparison for Image 1: input, original AGCC, improved AGCC, ground truth, histograms, and edges.



Figure 16. Phase 7 combined comparison grid for selected images.

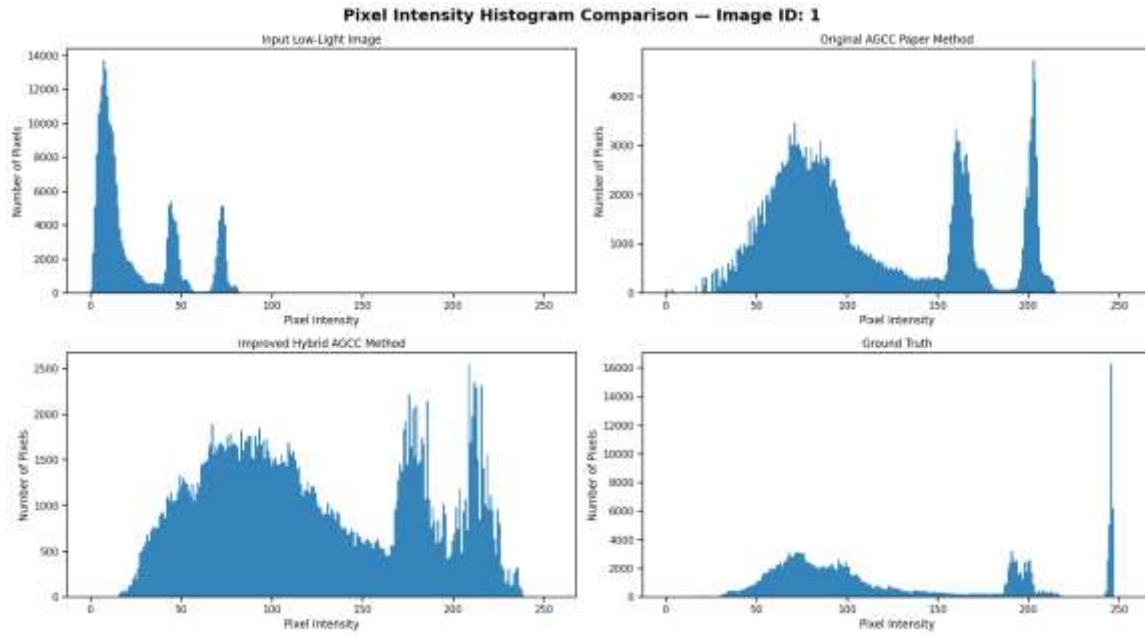


Figure 17. Final intensity histogram comparison for Image 1.

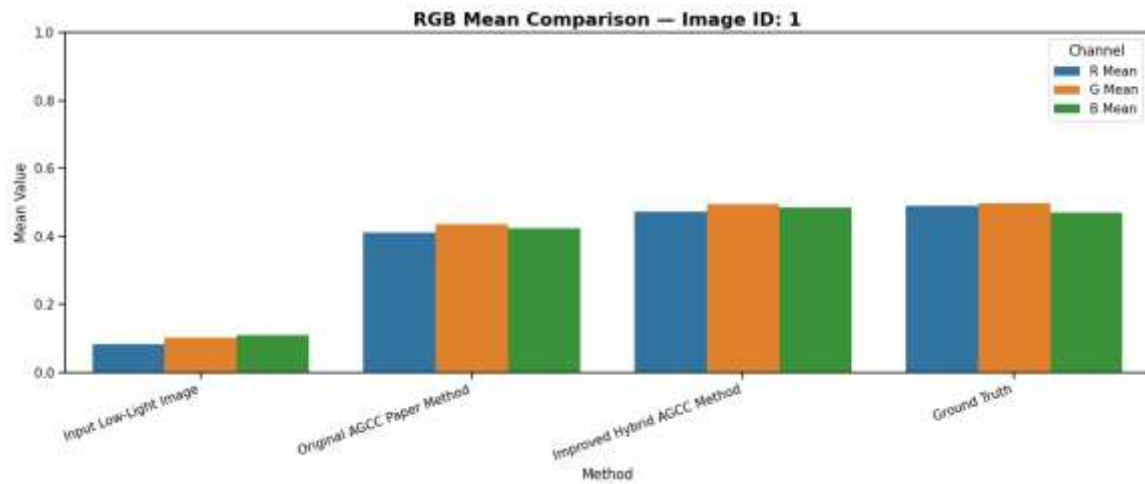


Figure 18. RGB mean comparison across methods.

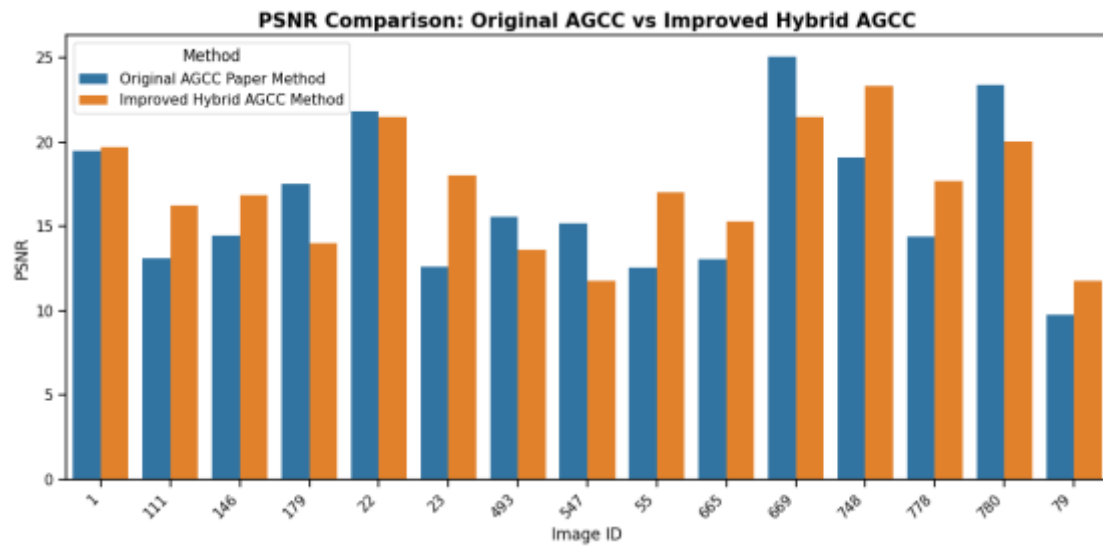


Figure 19. PSNR comparison: original AGCC vs improved hybrid AGCC.

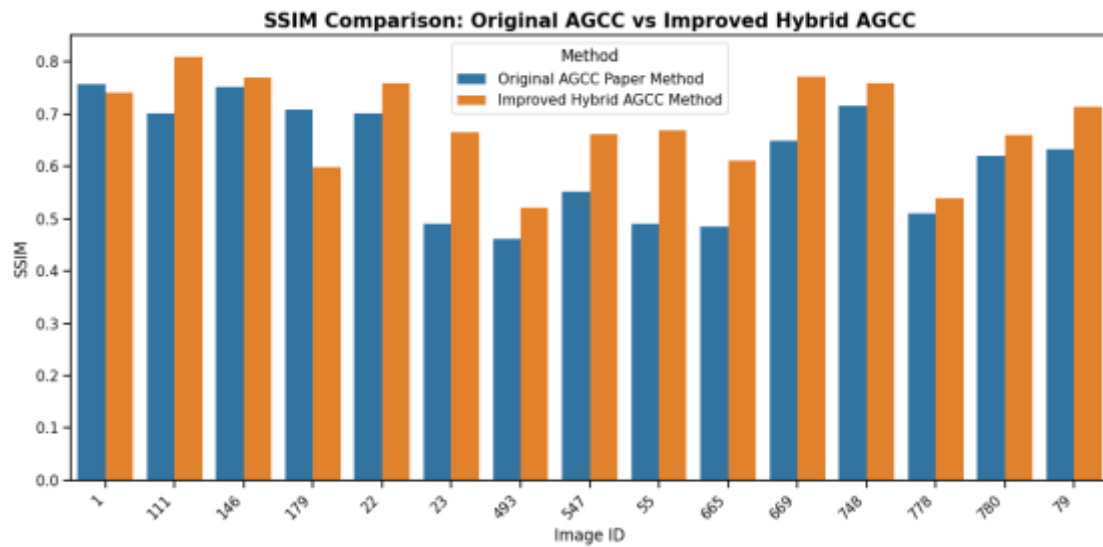


Figure 20. SSIM comparison: original AGCC vs improved hybrid AGCC.

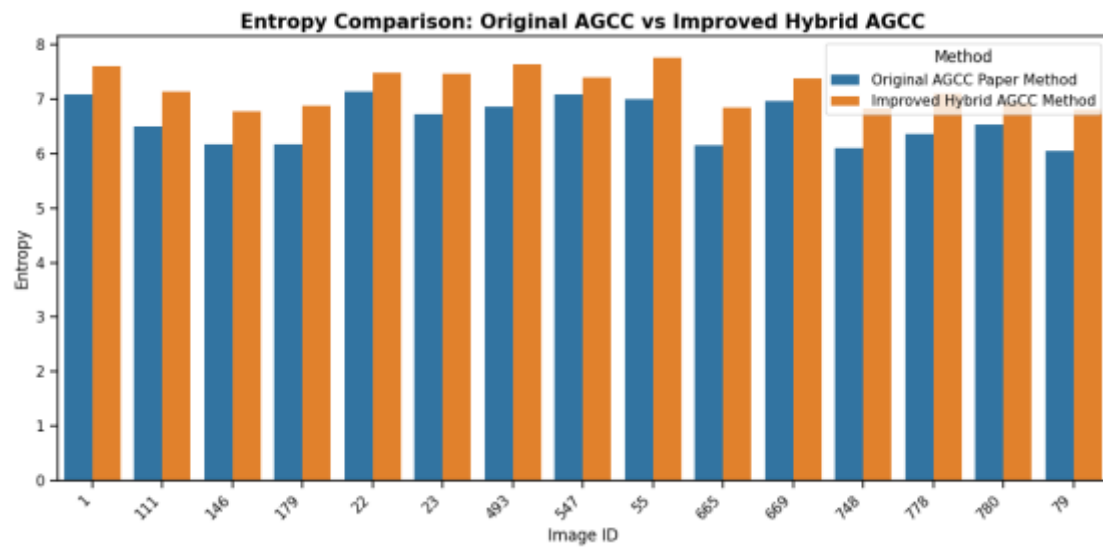


Figure 21. Entropy comparison: original AGCC vs improved hybrid AGCC.

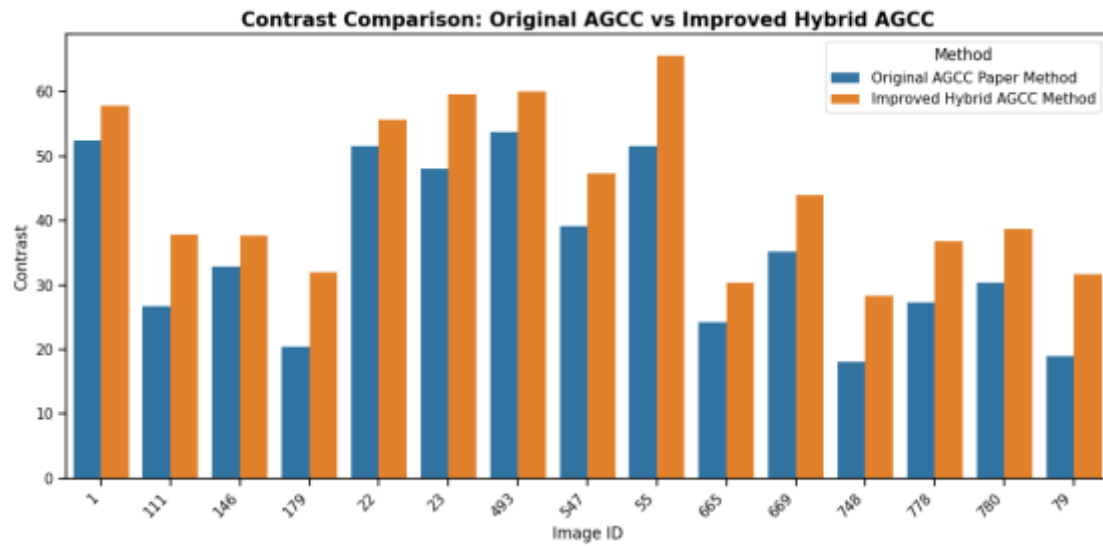


Figure 22. Contrast comparison: original AGCC vs improved hybrid AGCC.

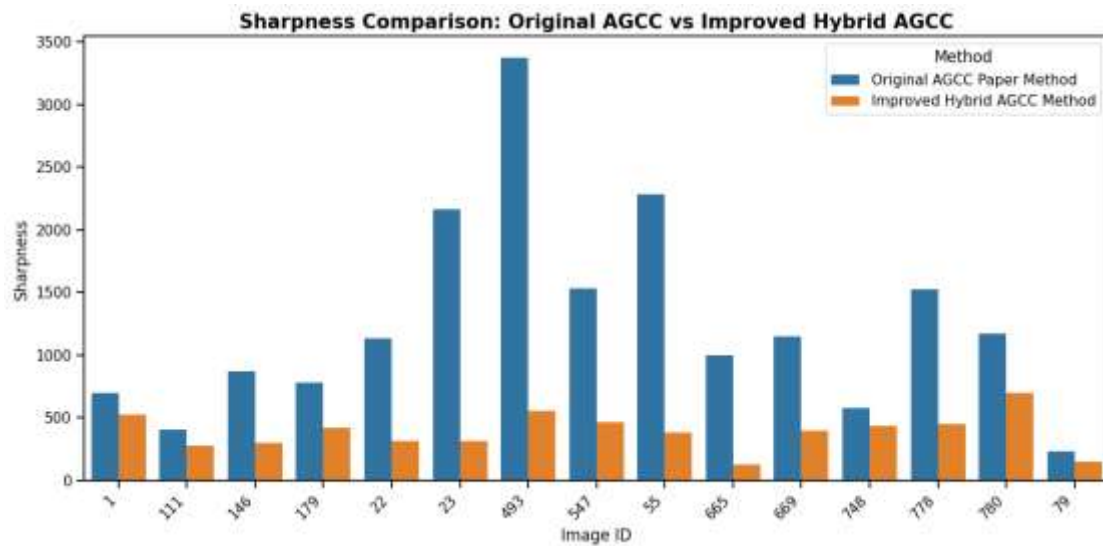


Figure 23. Sharpness comparison: original AGCC vs improved hybrid AGCC.

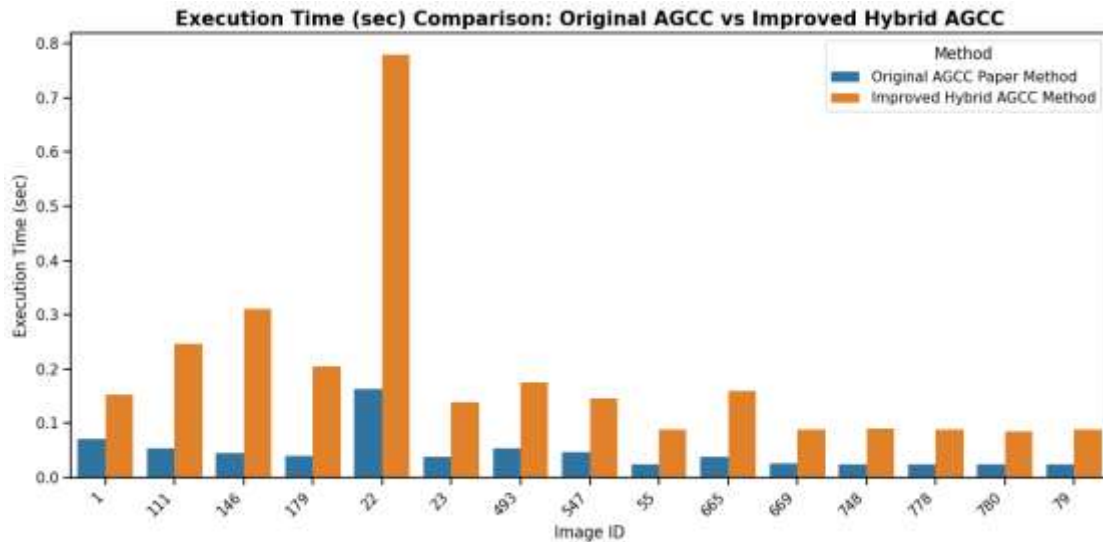


Figure 24. Execution time comparison: original AGCC vs improved hybrid AGCC.

9. Discussion

The results show that the original AGCC paper method is computationally efficient and produces strong enhancement with only three operations. Its strength is speed and simple color correction. The reproduced implementation confirmed that brightness and visual quality improve significantly compared with low-light input images. However, the reproduced method did not reach the exact paper-reported SSIM on the selected subset, mainly because dataset selection, image subset, person filtering, runtime environment, and implementation details differ from the original paper.

The Phase 6 and Phase 7 improvements demonstrate that additional DIP operations can improve several quality indicators. CLAHE increases local contrast, bilateral filtering reduces visual noise while preserving boundaries, and edge/morphology analysis provides stronger evidence about structural detail. The improved method produced higher PSNR, higher SSIM, lower NIQE, higher entropy, higher contrast, and higher intensity mean. These are meaningful improvements for the project criteria. At the same time, the improved method increases runtime and worsens BRISQUE and sharpness in the Phase 7 average table. This trade-off is important and should be explained during viva: enhancement methods can improve some metrics while degrading others because each metric measures a different property of the image.

10. Conclusion

This project successfully selected, analyzed, implemented, enhanced, and compared a 2024 low-light image enhancement research paper. The original AGCC method was reproduced using Python/OpenCV in Google Colab. Its three key stages - adaptive gamma correction, RGB mean-based color correction, and contrast stretching - were implemented and evaluated using PSNR, SSIM, NIQE, BRISQUE, runtime, histograms, and RGB statistics.

The proposed Phase 6 hybrid model extended AGCC using classical digital image processing techniques without deep learning. The Phase 7 comparative analysis showed that the improved hybrid method increased PSNR from 16.4673 to 17.2236, SSIM from 0.6151 to 0.6833, and improved NIQE from 7.5214 to 5.5877. It also improved entropy, contrast, and intensity mean. These improvements demonstrate that combining AGCC with local contrast enhancement and edge-preserving processing can produce a stronger enhanced result. The major limitation is increased runtime and mixed behavior in BRISQUE and sharpness

metrics. Overall, the project satisfies the required phases: paper analysis, technique identification, implementation, evaluation, improvement, and comparative analysis.

11. References

1. A. H. Alsaeedi, S. M. Hadi, and Y. Alazzawi, "Adaptive Gamma and Color Correction for Enhancing Low-Light Images," International Journal of Intelligent Engineering and Systems, vol. 17, no. 4, 2024.
2. G. Cao et al., "Contrast enhancement of brightness-distorted images by improved adaptive gamma correction," Computers & Electrical Engineering, vol. 66, pp. 569-582, 2018.
3. S. Hao et al., "Low-light image enhancement with semi-decoupled decomposition," IEEE Transactions on Multimedia, vol. 22, no. 12, pp. 3025-3038, 2020.
4. J. J. Jeon and J. Y. Park, "Low-light image enhancement using gamma correction prior in mixed color spaces," Pattern Recognition, vol. 146, 2024.
5. J. J. Jeon and I. K. Eom, "Low-light image enhancement using inverted image normalized by atmospheric light," Signal Processing, vol. 196, 2022.
6. M. Li et al., "Structure-revealing low-light image enhancement via robust Retinex model," IEEE Transactions on Image Processing, vol. 27, no. 6, pp. 2828-2841, 2018.
7. C. Wei et al., "Deep Retinex Decomposition for Low-Light Enhancement," LOL Dataset, 2018.
8. X. Guo et al., "LIME: Low-light image enhancement via illumination map estimation," IEEE Transactions on Image Processing, vol. 26, no. 2, pp. 982-993, 2016.
9. OpenCV Documentation, image processing library used for RGB conversion, filtering, CLAHE, edge detection, and morphology.
10. NumPy, Matplotlib, Scikit-Image, PyWavelets, and pyiqa documentation used for implementation, visualization, metric evaluation, and no-reference image quality assessment.

Appendix A: Submitted Code and Results

The submitted ZIP file includes all runnable notebooks, extracted outputs, metric CSV files, visual comparison grids, graphs, and input-output images. Full image-wise metric CSV files are included in the Data_Tables folder to avoid reducing readability of the main report with very wide tables.

Table 12. Final ZIP folder organization

Folder	Contents
Code	Runnable notebooks and clean Python implementation
Results/Phase4_Phase5	Original AGCC implementation outputs and histograms
Results/Phase6	Improved hybrid AGCC outputs, graphs, and comparison grids
Results/Phase7	Original vs improved comparative visuals and charts
Data Tables	All extracted CSV tables from executed notebooks
Presentation	10-15 minute presentation slide deck
References	Selected 2024 research paper PDF